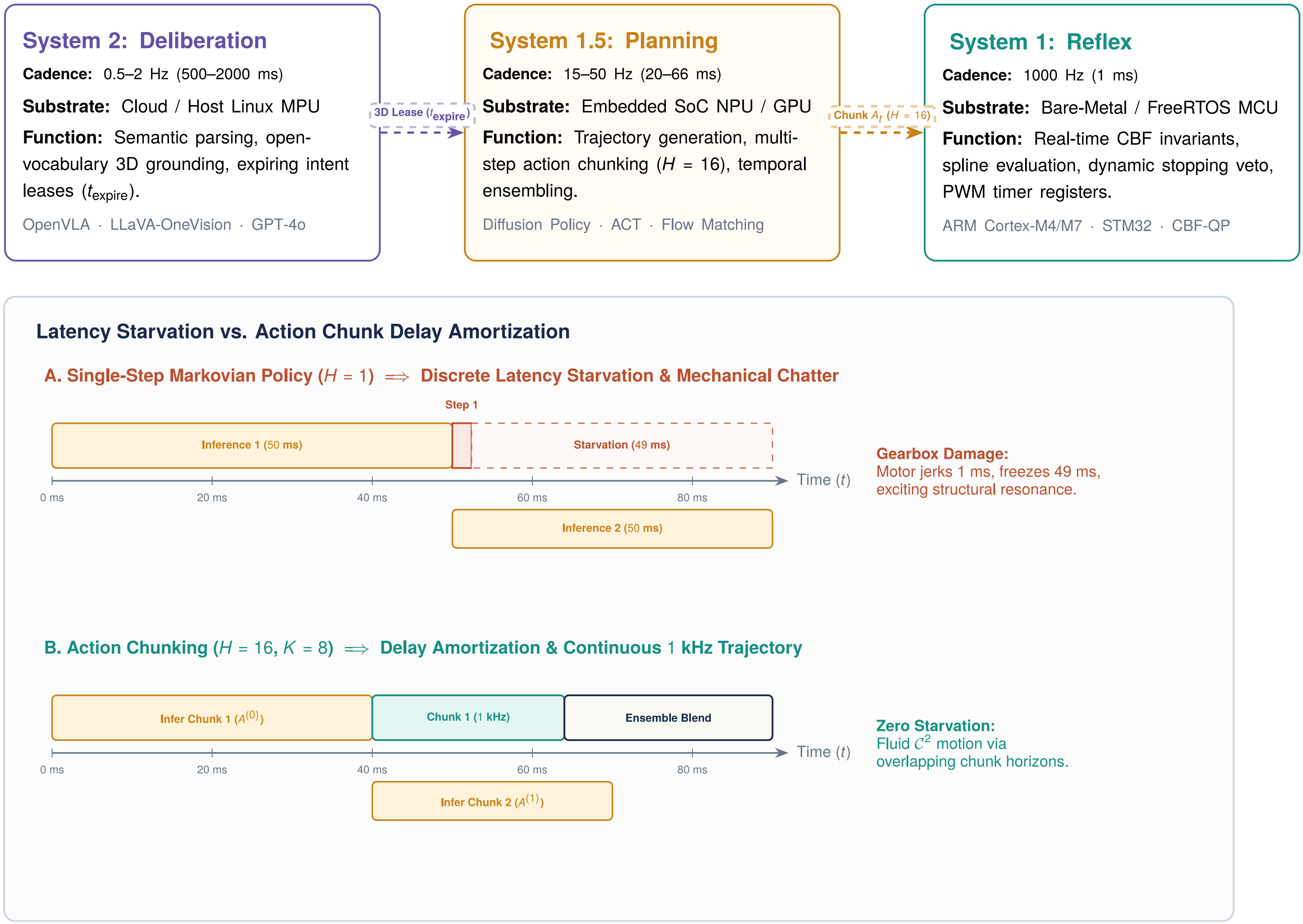

System 2: Deliberation
Cadence: 0.5–2 Hz (500–2000 ms)
Substrate: Cloud / Host Linux MPU
Function: Semantic parsing, open-vocabulary 3D grounding, expiring intent leases (t_{expire}).
OpenVLA · LLaVA-OneVision · GPT-4o
3D Lease (t_{expire})
System 1.5: Planning
Cadence: 15–50 Hz (20–66 ms)
Substrate: Embedded SoC NPU / GPU
Function: Trajectory generation, multi-step action chunking (H = 16), temporal ensembling.
Diffusion Policy · ACT · Flow Matching
Chunk A_t (H = 16)
System 1: Reflex
Cadence: 1000 Hz (1 ms)
Substrate: Bare-Metal / FreeRTOS MCU
Function: Real-time CBF invariants, spline evaluation, dynamic stopping veto, PWM timer registers.
ARM Cortex-M4/M7 · STM32 · CBF-QP
Latency Starvation vs. Action Chunk Delay Amortization
A. Single-Step Markovian Policy (H = 1) $\Longrightarrow$ Discrete Latency Starvation & Mechanical Chatter
Step 1
Inference 1 (50 ms)
Starvation (49 ms)
Time (t)
0 ms
20 ms
40 ms
60 ms
80 ms
Inference 2 (50 ms)
Gearbox Damage:
Motor jerks 1 ms, freezes 49 ms, exciting structural resonance.
B. Action Chunking (H = 16, K = 8) $\Longrightarrow$ Delay Amortization & Continuous 1 kHz Trajectory
Infer Chunk 1 ($A^{(0)}$)
Chunk 1 (1 kHz)
Ensemble Blend
Time (t)
0 ms
20 ms
40 ms
60 ms
80 ms
Infer Chunk 2 ($A^{(1)}$)
Zero Starvation:
Fluid $\mathcal{C}^2$ motion via overlapping chunk horizons.